SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour
Total Marks:30

QUESTIONS: 3

Annexure A

Dry Run Evaluation

Code of Conduct:

I Dharshan B 192312150 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Dharshan.

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1. Bidirectional search.

Given :-

Graph: $A \to B \to C \to D \to E \to F$

Start = A

Goal = F

Forward search:-

Starts from A and move toward F

$A \to B \to C \to D \to E \to F$

Backward search:-

Starts from F and move toward A.

$F \to E \to D \to C \to B \to A$

Expansion:-

Step	Forward from A	Backward from F
1	A	F
2	B	E
3	C	D
4	D	D

Meeting Point:-

Both Searches reach D.

Forward: $A \to B \to C \to D$.

Backward: $F \to E \to D$

Therefore,

Meeting Point = D.

Conclusion:-

Bidirectional search successfully finds the path from A to F by searching simultaneously from the start and goal.

2) Agent Architecture

A self-driving car receives information from camera, GPS, LiDAR and speed sensor. Based on this information, it controls the steering, brake and accelerator.

Sensors:-

1. camera:- Detects lane, traffic signals, vehicles and pedestrians
2. GPS:- Provides the current location and route information
3. LiDAR:- Detects obstacles and measures their distance
4. speed sensor:- Measures the current speed ~~of~~ the vehicle.

Actuators:-

- steering: controls the direction ~~of~~ the vehicle.
- Brake: slows down or stops the vehicle.
- Accelerator: Controls the vehicle's speed.

Suitable Agent:-

Model-Based utility-Based agent.

Justification:-

1. The model based agent maintains an internal representation of the environment using sensor information.
2. It can remember information about roads, obstacles, vehicles and traffic conditions.

3. The utility based component compares different possible action.

4. It can select an action based on safety, speed, comfort and efficiency.

3) CSP Exam Timetable.

Given subjects :- AI, ML, DBMS, OS

Available slots :-
1. Monday - 10 AM
2. Monday - 2 PM
3. Tuesday - 10 AM.

Step by step Backtracking

Step 1 : Assign AI

AI = Monday 10 AM.

Now DBMS must be after AI. So DBMS = Monday 2 PM or Tuesday 10 AM

Step 2 : Assign ML

ML cannot share the slot with AI.

∴ ML = Monday 2PM

Step 3 : Assign OS

ML must be before OS.

Since ML is on Monday 2PM :

OS = Tuesday 10 AM

This condition also satisfy.

step 4: Assign DBMS

DBMS must be after AI.

DBMS = Tuesday 10AM

Therefore, DBMS and OS can share Tuesday 10AM because there is no constraint preventing them from sharing.

Final Timetable.

Day & Time	Subject
Monday 10AM	AI
Monday 2 PM	ML
Tuesday 10AM	DBMS + OS.